



National University of Sciences & Technology (NUST)
School of Electrical Engineering and Computer Science (SEECS)
Department of Computing (DOC)

LEARN | EXPLORE | SOLVE | CREATE

*Mathematics
Drives
Computing
Forward*

Vector Algebra

CS-813 | Fall 2026

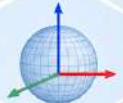
MSCS Program
SEECS, NUST

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$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$



$$A\mathbf{x} = \mathbf{b}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY



SEECS
SCHOOL OF ELECTRICAL ENGINEERING
AND COMPUTER SCIENCE



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A BRIGHTER TOMORROW

Euclidean Norm (Length of a Vector)

Definition:

Consider a d -dimensional vector

$\mathbf{x} = [x_1, x_2, \dots, x_d] \in \mathbb{R}^d$. The Euclidean norm (or length) of $\mathbf{x}$, denoted by $\|\mathbf{x}\|$, is the square root of the dot product of the vector with itself.

Expanded Form:

The squared norm is the sum of the squares of its components:

$$\|\mathbf{x}\|^2 = x_1^2 + x_2^2 + \dots + x_d^2 = \sum_{i=1}^d x_i^2$$

Numerical Example (3D):

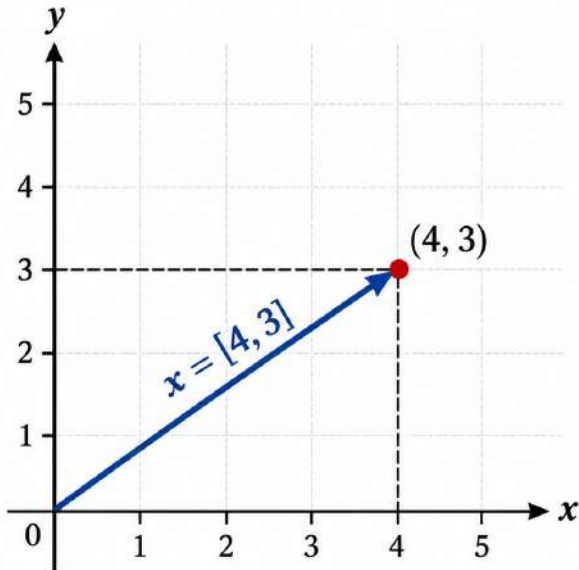
Let $\mathbf{x} = [2, -3, 6]$.

Step 1: Calculate the squared norm

(Sum of term-by-term products)

$$\begin{aligned}\|\mathbf{x}\|^2 &= (2 \cdot 2) + (-3 \cdot -3) + (6 \cdot 6) \\ &= 4 + 9 + 36 \\ &= 49\end{aligned}$$

Graphical Representation of $x = [4, 3]$



Step 1: Calculate the squared norm
(Sum of term-by-term products)

$$\begin{aligned}\|x\|^2 &= (4 \cdot 4) + (3 \cdot 3) \\ &= 16 + 9 \\ &= 25\end{aligned}$$

Step 2: Calculate the norm
(Take the square root)

$$\|x\| = \sqrt{25} = 5$$

The vector $x = [4, 3]$ has length 5,
i.e., $\|x\| = 5$.

Normalization of a Vector (Unit Vector)

The norm of a vector is the Euclidean distance of its coordinates from the origin. It represents the physical length of the directed line segment.

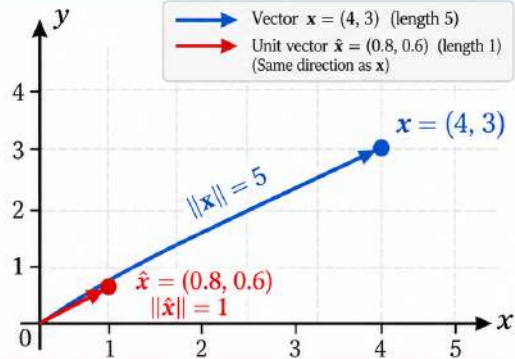
Normalization (Unit Vector):

Often, vectors are normalized to unit length by dividing them with their norm. This process scales the vector so its magnitude becomes exactly 1, while preserving its original direction.

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|} = \frac{\mathbf{x}}{\sqrt{\mathbf{x} \cdot \mathbf{x}}}$$

Key Properties:

- The resulting vector $\hat{\mathbf{x}}$ is called a **unit vector**.
- It has a magnitude of 1: $\|\hat{\mathbf{x}}\| = 1$.
- Normalization only changes the magnitude, not the direction.



Example:

For $\mathbf{x} = [4, 3]$,

$$\|\mathbf{x}\| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$$

$$\hat{\mathbf{x}} = \frac{1}{5} [4, 3] = [0.8, 0.6]$$

The unit vector points in the **same** direction as $\mathbf{x}$, but has length 1.

Angle Between Two Vectors

For two non-zero vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^d$, the angle θ between them is defined using the dot product:

$$\cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

Key Properties:

- ▶ $\theta \in [0, \pi]$ is the smaller angle between $\mathbf{x}$ and $\mathbf{y}$.
- ▶ If $\mathbf{x} \cdot \mathbf{y} > 0$, then $0 \leq \theta < 90^\circ$ (acute angle).
- ▶ If $\mathbf{x} \cdot \mathbf{y} = 0$, then $\theta = 90^\circ$ (orthogonal vectors).
- ▶ If $\mathbf{x} \cdot \mathbf{y} < 0$, then $90^\circ < \theta \leq 180^\circ$ (obtuse angle).

Angle Between Two Vectors: Numerical Example

Example:

Find the angle θ between the vectors $\mathbf{a} = [4, 3]$ and $\mathbf{b} = [1, 2]$.

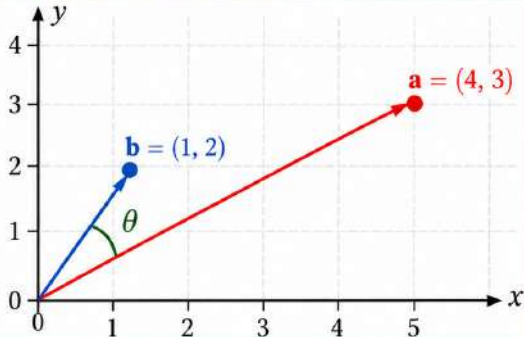
Formula:

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad \theta = \cos^{-1} \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \right)$$

Step-by-step Solution:

- 1 Compute the dot product:
 $\mathbf{a} \cdot \mathbf{b} = 4(1) + 3(2) = 4 + 6 = 10$
- 2 Find the norms:
 $\|\mathbf{a}\| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5 \quad \|\mathbf{b}\| = \sqrt{1^2 + 2^2} = \sqrt{5}$
- 3 Compute the cosine of the angle:
 $\cos \theta = \frac{10}{5\sqrt{5}} = \frac{2}{\sqrt{5}} \approx 0.8944$
- 4 Find the angle: $\theta = \cos^{-1}(0.8944) \approx 26.6^\circ$

Graphical Representation:



Result:

The angle between $\mathbf{a} = [4, 3]$ and $\mathbf{b} = [1, 2]$ is

$$\theta \approx 26.6^\circ$$

Orthogonal Vectors

A pair of vectors is **orthogonal** if their dot product is 0, and the angle between them is 90° (for non-zero vectors).

$$\mathbf{x} \cdot \mathbf{y} = 0 \iff \theta = 90^\circ$$

Special Case:

- The zero vector $\mathbf{0}$ is considered orthogonal to every vector.

Example:

Let $\mathbf{x} = [2, -3]$ and $\mathbf{y} = [3, 2]$.

Calculate the dot product:

$$\begin{aligned}\mathbf{x} \cdot \mathbf{y} &= (2 \cdot 3) + (-3 \cdot 2) \\ &= 6 - 6 = 0.\end{aligned}$$

Since the dot product is 0, vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal.

Example: Orthogonal Vectors (Dot Product = 0)

Given:

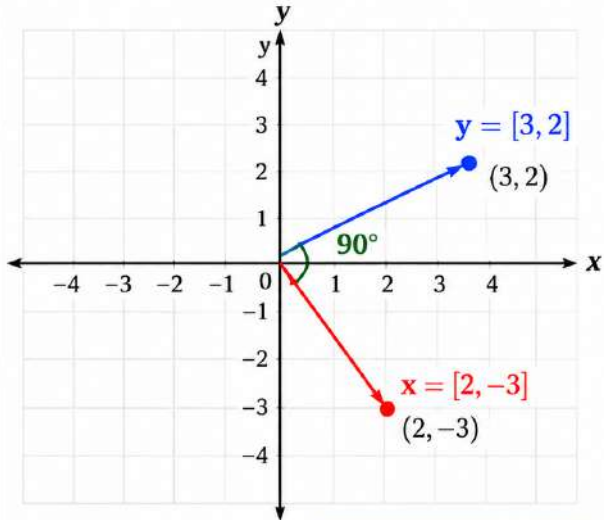
Let $\mathbf{x} = [2, -3]$ and $\mathbf{y} = [3, 2]$.

Calculate the dot product:

$$\begin{aligned}\mathbf{x} \cdot \mathbf{y} &= (2 \cdot 3) + (-3 \cdot 2) \\ &= 6 - 6 \\ &= 0.\end{aligned}$$

Conclusion:

Since the dot product is 0, the vectors $\mathbf{x}$ and $\mathbf{y}$ are **orthogonal** (perpendicular).



$$\theta = 90^\circ \Rightarrow \mathbf{x} \cdot \mathbf{y} = 0$$

Set of Orthogonal Vectors

A set of vectors is **mutually orthogonal** if every distinct pair of vectors in the set has a dot product of 0.

$$\mathbf{x}_i \cdot \mathbf{x}_j = 0 \quad \text{for all } i \neq j$$

► Any set containing the zero vector $\mathbf{0}$ is orthogonal, as $\mathbf{0} \cdot \mathbf{v} = 0$ for any vector $\mathbf{v}$.

Example:

Let $\mathbf{a} = [1, 1, 0]$, $\mathbf{b} = [1, -1, 0]$, and $\mathbf{c} = [0, 0, 2]$.

Check pairwise dot products:

$$\mathbf{a} \cdot \mathbf{b} = (1 \cdot 1) + (1 \cdot -1) + (0 \cdot 0) = 1 - 1 + 0 = 0$$

$$\mathbf{a} \cdot \mathbf{c} = (1 \cdot 0) + (1 \cdot 0) + (0 \cdot 2) = 0 + 0 + 0 = 0$$

$$\mathbf{b} \cdot \mathbf{c} = (1 \cdot 0) + (-1 \cdot 0) + (0 \cdot 2) = 0 + 0 + 0 = 0$$

Since every pair has a dot product of 0, the set $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ is a set of mutually orthogonal vectors.

Orthonormal Vectors

A pair of vectors is **orthonormal** if they are orthogonal to each other and each vector has a norm (length) of 1.

$$\mathbf{x} \cdot \mathbf{y} = 0 \quad \text{and} \quad \|\mathbf{x}\| = \|\mathbf{y}\| = 1$$

► The standard basis vectors $\mathbf{e}_1 = [1, 0]$ and $\mathbf{e}_2 = [0, 1]$ in $\mathbb{R}^2$ are orthonormal.

Example:

Let $\mathbf{x} = [\frac{3}{5}, \frac{4}{5}]$ and $\mathbf{y} = [-\frac{4}{5}, \frac{3}{5}]$. **Check orthogonality yourself!**

$$\|\mathbf{x}\| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{25}{25}} = 1$$

$$\|\mathbf{y}\| = \sqrt{\left(-\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \sqrt{\frac{16}{25} + \frac{9}{25}} = \sqrt{\frac{25}{25}} = 1$$

Example: Orthonormal Vectors

Given:

$$\mathbf{x} = \left[\frac{3}{5}, \frac{4}{5} \right] \quad \text{and} \quad \mathbf{y} = \left[-\frac{4}{5}, \frac{3}{5} \right].$$

1. Check the norms:

$$\|\mathbf{x}\| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{25}{25}} = 1$$

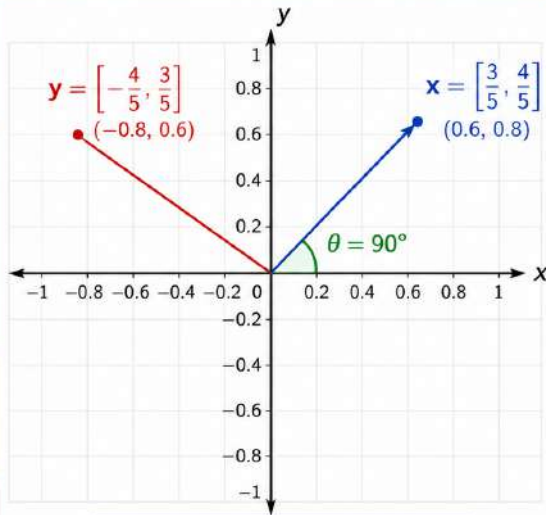
$$\|\mathbf{y}\| = \sqrt{\left(-\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \sqrt{\frac{16}{25} + \frac{9}{25}} = \sqrt{\frac{25}{25}} = 1$$

2. Check the dot product:

$$\mathbf{x} \cdot \mathbf{y} = \left(\frac{3}{5}\right)\left(-\frac{4}{5}\right) + \left(\frac{4}{5}\right)\left(\frac{3}{5}\right) = -\frac{12}{25} + \frac{12}{25} = 0$$

Conclusion:

Since both norms are 1 and the dot product is 0, the vectors $\{\mathbf{x}, \mathbf{y}\}$ are **orthonormal**.



**Unit vectors at right angles
(orthonormal pair)**

Set of Orthonormal Vectors

A set of vectors is **orthonormal** if every distinct pair in the set is orthogonal and the norm of each vector is 1.

$$\mathbf{v}_i \cdot \mathbf{v}_j = 0 \quad \text{for all } i \neq j \quad \text{and} \quad \|\mathbf{v}_i\| = 1 \quad \text{for all } i$$

► The standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ in $\mathbb{R}^3$ is a classic orthonormal set.

Example:

Let $\mathbf{u}_1 = [1, 0, 0]$, $\mathbf{u}_2 = [0, 1, 0]$, and $\mathbf{u}_3 = [0, 0, 1]$.

Check norms:

$$\|\mathbf{u}_1\| = \sqrt{1^2 + 0^2 + 0^2} = 1$$

$$\|\mathbf{u}_2\| = \sqrt{0^2 + 1^2 + 0^2} = 1$$

$$\|\mathbf{u}_3\| = \sqrt{0^2 + 0^2 + 1^2} = 1$$

Check pairwise orthogonality:

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = (1 \cdot 0) + (0 \cdot 1) + (0 \cdot 0) = 0$$

$$\mathbf{u}_1 \cdot \mathbf{u}_3 = (1 \cdot 0) + (0 \cdot 0) + (0 \cdot 1) = 0$$

$$\mathbf{u}_2 \cdot \mathbf{u}_3 = (0 \cdot 0) + (1 \cdot 0) + (0 \cdot 1) = 0$$

Example: Orthonormal Vectors in $\mathbb{R}^3$ (Standard Basis)

Given:

$$\mathbf{u}_1 = [1, 0, 0], \quad \mathbf{u}_2 = [0, 1, 0], \quad \mathbf{u}_3 = [0, 0, 1].$$

Check norms:

$$\|\mathbf{u}_1\| = \sqrt{1^2 + 0^2 + 0^2} = 1$$

$$\|\mathbf{u}_2\| = \sqrt{0^2 + 1^2 + 0^2} = 1$$

$$\|\mathbf{u}_3\| = \sqrt{0^2 + 0^2 + 1^2} = 1$$

Check pairwise orthogonality:

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = (1 \cdot 0) + (0 \cdot 1) + (0 \cdot 0) = 0$$

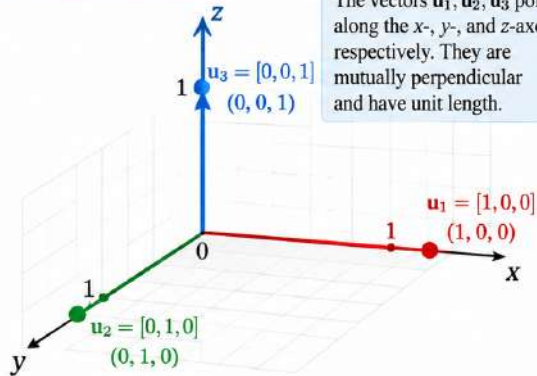
$$\mathbf{u}_1 \cdot \mathbf{u}_3 = (1 \cdot 0) + (0 \cdot 0) + (0 \cdot 1) = 0$$

$$\mathbf{u}_2 \cdot \mathbf{u}_3 = (0 \cdot 0) + (1 \cdot 0) + (0 \cdot 1) = 0$$

Conclusion:

Since all norms are 1 and all pairwise dot products are 0, the set $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is **orthonormal**.

3D Representation



The vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ point along the x-, y-, and z-axes respectively. They are mutually perpendicular and have unit length.

Standard basis vectors in $\mathbb{R}^3$ form an orthonormal set.

1 Vectors in $\mathbb{R}^3$

Let $\mathbf{a} = (1, 2, -1)$ and $\mathbf{b} = (2, -1, 2)$ in $\mathbb{R}^3$.

2 Dot Product

For $\mathbf{a} = (1, 2, -1)$ and $\mathbf{b} = (2, -1, 2)$,

$$\begin{aligned}\mathbf{a} \cdot \mathbf{b} &= 1(2) + 2(-1) + (-1)(2) \\ &= 2 - 2 - 2 = -2.\end{aligned}$$

Since $\mathbf{a} \cdot \mathbf{b} \neq 0$, the vectors are not orthogonal.

3 Norms (Lengths)

$$\|\mathbf{a}\| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}$$

$$\|\mathbf{b}\| = \sqrt{2^2 + (-1)^2 + 2^2} = 3$$

4 Euclidean Distance

$$\begin{aligned}\|\mathbf{a} - \mathbf{b}\| &= \sqrt{(1-2)^2 + (2-(-1))^2 + (-1-2)^2} \\ &= \sqrt{(-1)^2 + 3^2 + (-3)^2} = \sqrt{19}.\end{aligned}$$

5 Angle Between Vectors

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{-2}{\sqrt{6} \times 3} = \frac{-2}{3\sqrt{6}}$$

$$\theta = \cos^{-1} \left(\frac{-2}{3\sqrt{6}} \right) \approx 106.26^\circ.$$

The angle between $\mathbf{a}$ and $\mathbf{b}$ is approximately 106.26° , which is obtuse (greater than 90°).

6 Orthogonal Vectors (Example)

Let $\mathbf{u} = (1, 0, 1)$ and $\mathbf{v} = (1, 0, -1)$.

$$\mathbf{u} \cdot \mathbf{v} = 1(1) + 0(0) + 1(-1) = 0.$$

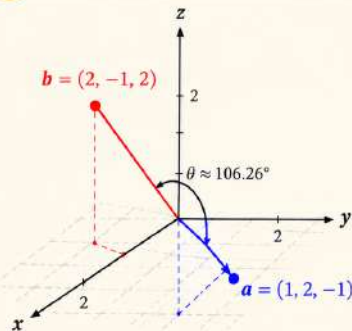
Hence, $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors.

7 Orthonormal Vectors (Example)

Let $\mathbf{e}_1 = (1, 0, 0)$, $\mathbf{e}_2 = (0, 1, 0)$ and $\mathbf{e}_3 = (0, 0, 1)$.

- $\mathbf{e}_i \cdot \mathbf{e}_j = 0$ for $i \neq j$.
- $\|\mathbf{e}_1\| = \|\mathbf{e}_2\| = \|\mathbf{e}_3\| = 1$.
- Thus, $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is an orthonormal set in $\mathbb{R}^3$ (the standard basis).

8 Geometric Interpretation in $\mathbb{R}^3$



NLP Document Similarity

In NLP, document vectors are compared using the angle θ and Cosine Similarity.

1. Not Similar (Orthogonal Vectors)

When two documents are "not similar," they share no common words, topics, or contextual meaning. They exist in completely different semantic spaces.

Mathematical Properties:

- ▶ **Cosine Similarity:** 0
- ▶ **Angle (θ):** 90° (Orthogonal)
- ▶ **Dot Product:** $\mathbf{x} \cdot \mathbf{y} = 0$

NLP Example:

- ▶ **Document A:** "Quantum Physics"
- ▶ **Document B:** "Baking a Chocolate Cake"

These documents are completely unrelated, but they are not opposites.

NLP Document Similarity (Entirely Opposite Vectors)

When two documents are "entirely opposite," they represent directly contradictory concepts or sentiments. The vectors point in exactly opposite directions in the semantic space.

Cosine Similarity: -1 **Angle (θ):** 180° **Dot Product:** $\mathbf{x} \cdot \mathbf{y} < 0$ (highly negative)

NLP Example (Word Embeddings): In models like Word2Vec, words are mapped to vectors. Consider the vector for "happy" and "sad":

► **Vector for "happy":** $\mathbf{v}_{\text{happy}} = [1, 0]$

► **Vector for "sad":** $\mathbf{v}_{\text{sad}} = [-1, 0]$

Calculate Cosine Similarity:

$$\cos \theta = \frac{[1, 0] \cdot [-1, 0]}{\| [1, 0] \| \| [-1, 0] \|} = \frac{-1}{1 \cdot 1} = -1$$

Since the Cosine Similarity is -1 , the vectors point in exactly opposite directions.

NLP Document Similarity

Summary of Vector Relationships

Relationship	Angle (θ)	Cosine Sim.	NLP Interpretation
Highly Similar	0°	1	Same topic, same sentiment
Not Similar	90°	0	Unrelated topics
Entirely Opposite	180°	-1	Directly contradictory meaning

Key Takeaway:

"Not similar" means the vectors are perpendicular (no shared meaning). "Entirely opposite" means the vectors are collinear but pointing in opposite directions (directly contradictory meaning).

Application: Cosine Similarity in NLP

Measuring Semantic Similarity Between Documents

Problem:

We want to measure how similar the meanings of two short documents are using cosine similarity.

Document 1 (d_1):

“cat animal pet”

Document 2 (d_2):

“dog animal pet”

Step 1: Build a vocabulary

Vocabulary (unique words):
{cat, dog, animal, pet}

Step 2: Vectorize the documents (Bag-of-Words)

Represent each document by the count of each word in the vocabulary.

Step 3: Document vectors

Word	d_1	d_2
cat	1	0
dog	0	1
animal	1	1
pet	1	1

So,

$$\mathbf{d}_1 = [1, 0, 1, 1] \quad \mathbf{d}_2 = [0, 1, 1, 1]$$

Step 4: Compute cosine similarity

$$\cos(\theta) = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{\|\mathbf{d}_1\| \|\mathbf{d}_2\|}$$

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = 1(0) + 0(1) + 1(1) + 1(1) = 2$$

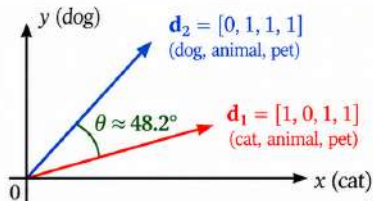
$$\|\mathbf{d}_1\| = \sqrt{1^2 + 0^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\|\mathbf{d}_2\| = \sqrt{0^2 + 1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\cos(\theta) = \frac{2}{\sqrt{3} \cdot \sqrt{3}} = \frac{2}{3} \approx 0.667$$

$$\theta = \cos^{-1}(0.667) \approx 48.2^\circ$$

Graphical Representation (2D Illustration)



Interpretation:

$\cos(\theta) \approx 0.667$ (moderate to high similarity)

The two documents share common words (animal, pet), so they are semantically **related**, even though they have different specific words (cat vs. dog).

Angle-based Condition for Similarity:

Angle θ	$\cos(\theta)$	Meaning
$\theta = 0^\circ$	1	Exactly the same (identical)
$0^\circ < \theta < 90^\circ$	$0 < \cos(\theta) < 1$	Similar (more similar as $\theta \rightarrow 0^\circ$)
$\theta = 90^\circ$	0	Not similar (orthogonal)
$90^\circ < \theta < 180^\circ$	$-1 < \cos(\theta) < 0$	Dissimilar (opposite in meaning)
$\theta = 180^\circ$	-1	Completely opposite

Euclidean Distance (Squared)

Consider two d -dimensional vectors $\mathbf{x} = [x_1, x_2, \dots, x_d]$ and $\mathbf{y} = [y_1, y_2, \dots, y_d]$:

Euclidean Distance: (squared)

The (squared) Euclidean distance between them can be shown to be the dot product of $\mathbf{x} - \mathbf{y}$ with itself:

$$\|\mathbf{x} - \mathbf{y}\|^2 = (\mathbf{x} - \mathbf{y}) \cdot (\mathbf{x} - \mathbf{y}) = \sum_{i=1}^d |x_i - y_i|^2$$

The norm of the vector is the Euclidean distance of its coordinates from the origin.

Example (2D): Difference of Two Vectors

Given:

Let $\mathbf{x} = [5, 8]$ and $\mathbf{y} = [2, 4]$.

Compute the difference:

$$\mathbf{x} - \mathbf{y} = [5 - 2, 8 - 4] = [3, 4].$$

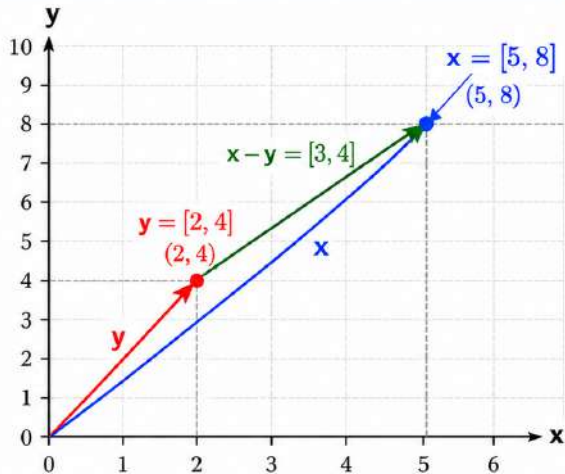
Compute the norm:

$$\begin{aligned}\|\mathbf{x} - \mathbf{y}\|^2 &= 3^2 + 4^2 \\ &= 9 + 16 \\ &= 25\end{aligned}$$

$$\|\mathbf{x} - \mathbf{y}\| = \sqrt{25} = 5.$$

Result:

$$\mathbf{x} - \mathbf{y} = [3, 4] \text{ and } \|\mathbf{x} - \mathbf{y}\| = 5.$$



The vector $\mathbf{x} - \mathbf{y} = [3, 4]$ is drawn **from the tip of $\mathbf{y} = [2, 4]$ to the tip of $\mathbf{x} = [5, 8]$** , and has length 5.

Example (3D): Euclidean Distance Between Two Points

Given:

Let $P = (1, 2, 3)$ and $Q = (4, -1, 2)$ in $\mathbb{R}^3$.

Distance formula in 3D:

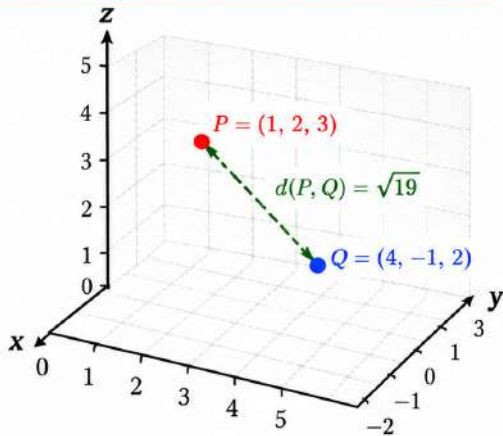
$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Compute the distance:

$$\begin{aligned} d(P, Q) &= \sqrt{(4 - 1)^2 + (-1 - 2)^2 + (2 - 3)^2} \\ &= \sqrt{3^2 + (-3)^2 + (-1)^2} \\ &= \sqrt{9 + 9 + 1} \\ &= \sqrt{19} \\ &\approx 4.36 \end{aligned}$$

Result:

The Euclidean distance between P and Q is $d(P, Q) = \sqrt{19} \approx 4.36$.



The Euclidean distance $d(P, Q) = \sqrt{19}$ is the length of the line segment joining the points P and Q in 3D space.

Application of Euclidean Distance: k-Nearest Neighbours

k-NN is a simple, non-parametric classification algorithm. It classifies a new data point based on the majority label of its k closest neighbours in the training set. Euclidean distance is the most commonly used metric to measure "closeness".

Real-World Applications:

- ▶ Handwritten Digit Recognition
- ▶ Recommendation Systems
- ▶ Medical Diagnosis
- ▶ Spam Filtering

How Euclidean Distance is Used:

Step 1: Represent each data point as a vector in $\mathbb{R}^d$. **Step 2:** For a new point $\mathbf{z}$, calculate $\|\mathbf{z} - \mathbf{x}_i\|$ for all training points.

Step 3: Sort distances and select the k smallest. **Step 4:** Assign the majority class label among those k neighbours.

Application: Nearest Neighbour Classification (2D)

Example:

Suppose we have two classes of training points:

Class A (Red): $\mathbf{p}_1 = [1, 1]$, $\mathbf{p}_2 = [2, 3]$ **Class B (Blue):** $\mathbf{q}_1 = [6, 5]$, $\mathbf{q}_2 = [7, 7]$

A new point $\mathbf{z} = [3, 4]$ needs to be classified.

Distance to Class A:

$$\begin{aligned}\|\mathbf{z} - \mathbf{p}_1\| &= \sqrt{(3-1)^2 + (4-1)^2} \\ &= \sqrt{4+9} = \sqrt{13} \approx 3.61\end{aligned}$$

$$\begin{aligned}\|\mathbf{z} - \mathbf{p}_2\| &= \sqrt{(3-2)^2 + (4-3)^2} \\ &= \sqrt{1+1} = \sqrt{2} \approx 1.41\end{aligned}$$

Distance to Class B:

$$\begin{aligned}\|\mathbf{z} - \mathbf{q}_1\| &= \sqrt{(3-6)^2 + (4-5)^2} \\ &= \sqrt{9+1} = \sqrt{10} \approx 3.16\end{aligned}$$

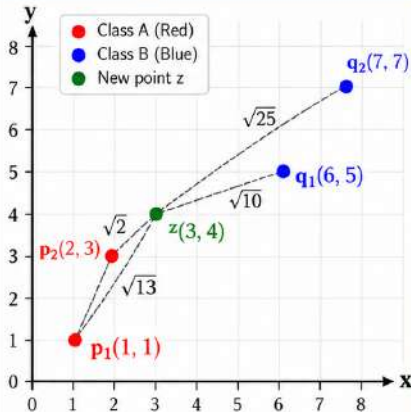
$$\begin{aligned}\|\mathbf{z} - \mathbf{q}_2\| &= \sqrt{(3-7)^2 + (4-7)^2} \\ &= \sqrt{16+9} = \sqrt{25} = 5.00\end{aligned}$$

Conclusion:

The smallest distance is $\sqrt{2} \approx 1.41$, which corresponds to $\mathbf{p}_2$ from Class A.

Therefore, the new point $\mathbf{z} = [3, 4]$ is classified as **Class A (Red)**.

Training Points and New Point



Manhattan Distance

Consider two d -dimensional vectors $\mathbf{x} = [x_1, x_2, \dots, x_d]$ and $\mathbf{y} = [y_1, y_2, \dots, y_d]$:

Definition:

The Manhattan distance (also called L_1 norm or Taxicab distance) is the sum of the absolute differences of their coordinates. It measures distance along grid lines, similar to navigating city blocks.

$$d_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 = \sum_{i=1}^d |x_i - y_i|$$

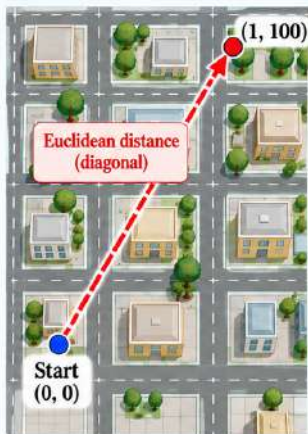
Comparison with Euclidean:

- ▶ **Euclidean (L_2):** Straight-line distance, $\sqrt{\sum (x_i - y_i)^2}$
- ▶ **Manhattan (L_1):** Grid-based distance, $\sum |x_i - y_i|$
- ▶ Manhattan distance is more robust to outliers.

Euclidean vs. Manhattan (Taxicab) Distance

Euclidean Distance

(Straight-line / Diagonal Distance)



Euclidean distance:

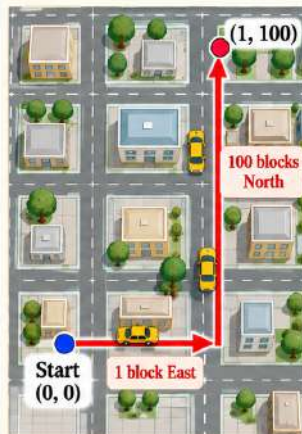
$$d_E = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\begin{aligned} d_E &= \sqrt{1^2 + 100^2} \\ &= \sqrt{10001} \\ &\approx 100.00 \end{aligned}$$

This is the straight-line distance (as if you could fly diagonally).

Manhattan Distance

(Taxicab Distance – along the grid)



Manhattan distance:

$$d_M = |x_2 - x_1| + |y_2 - y_1|$$

$$\begin{aligned} d_M &= |1 - 0| + |100 - 0| \\ &= 1 + 100 \\ &= 101 \end{aligned}$$

This is the distance you walk due to the city grid (like a taxi), adding the lengths of the horizontal and vertical moves.

Manhattan Distance: Numerical Example

$$d_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 = \sum_{i=1}^d |x_i - y_i|$$

Example (2D):

Let $\mathbf{x} = [5, 2]$ and $\mathbf{y} = [1, 6]$.

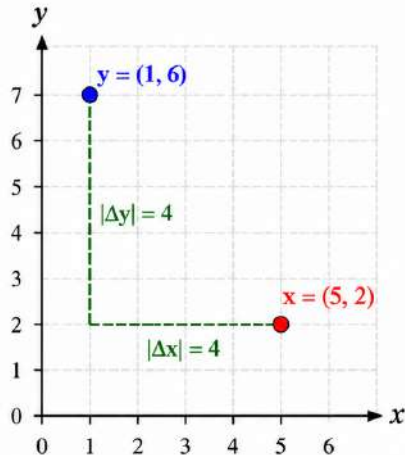
Step 1: Compute the absolute differences

$$|x_1 - y_1| = |5 - 1| = 4$$

$$|x_2 - y_2| = |2 - 6| = |-4| = 4$$

Step 2: Sum the absolute differences

$$\begin{aligned} d_1(\mathbf{x}, \mathbf{y}) &= |5 - 1| + |2 - 6| \\ &= 4 + 4 \\ &= 8 \end{aligned}$$



Result: The Manhattan distance between $\mathbf{x}$ and $\mathbf{y}$ is 8.

Applications of Manhattan Distance in ML

Lasso Regression (L_1 Regularization): Adds a penalty equal to the absolute value of coefficients.

Robust k-Nearest Neighbours (k-NN): Used instead of Euclidean distance when data contains outliers.

High-Dimensional Data (NLP & Text): More stable than Euclidean distance in sparse, high-dimensional spaces (e.g., document similarity) where the "Curse of Dimensionality" affects other metrics.

Image Processing & Computer Vision: It is less sensitive to noise and extreme pixel values compared to Euclidean distance.

Recommendation Systems: Handles sparse user-item matrices and extreme ratings robustly when finding similar users or items. **K-Medians Clustering:** Uses Manhattan distance and medians instead of means. It is computationally efficient and robust to outliers in large datasets.

k -NN Classification using Manhattan Distance

Problem Setup:

Suppose we have two classes of training points in $\mathbb{R}^2$:

- **Class A (Red):** $\mathbf{p}_1 = [1, 1]$, $\mathbf{p}_2 = [2, 3]$
- **Class B (Blue):** $\mathbf{q}_1 = [6, 5]$, $\mathbf{q}_2 = [7, 7]$

A new point $\mathbf{z} = [3, 4]$ needs to be classified using $k = 1$ (nearest neighbour).

Step 1: Compute Manhattan distances

Distance to Class A:

$$\begin{aligned}d_1(\mathbf{z}, \mathbf{p}_1) &= |3 - 1| + |4 - 1| \\&= 2 + 3 = 5\end{aligned}$$

$$\begin{aligned}d_1(\mathbf{z}, \mathbf{p}_2) &= |3 - 2| + |4 - 3| \\&= 1 + 1 = 2\end{aligned}$$

Distance to Class B:

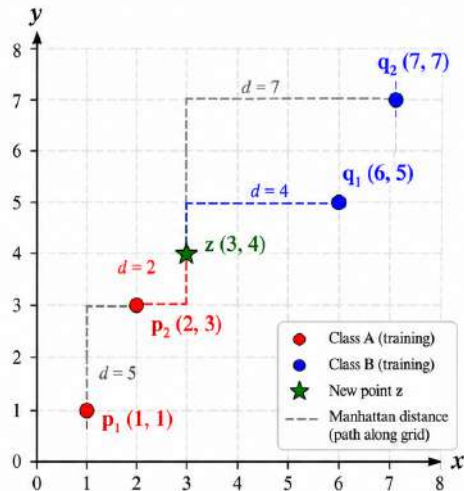
$$\begin{aligned}d_1(\mathbf{z}, \mathbf{q}_1) &= |3 - 6| + |4 - 5| \\&= 3 + 1 = 4\end{aligned}$$

$$\begin{aligned}d_1(\mathbf{z}, \mathbf{q}_2) &= |3 - 7| + |4 - 7| \\&= 4 + 3 = 7\end{aligned}$$

Step 2: Find the minimum distance

The smallest Manhattan distance is 2, which corresponds to $\mathbf{p}_2$ from Class A.

Conclusion: The new point $\mathbf{z} = [3, 4]$ is classified as Class A.



Geometric Interpretation of Orthogonal Projection

Geometrically, orthogonal projection represents the "**shadow**" or "**footprint**" that one vector casts onto another when light is shone perpendicular to it.

1. The "Shadow" Analogy

Imagine vector $\mathbf{x}$ is a flat stick lying on the ground, and vector $\mathbf{y}$ is a second stick pointing up into the air at an angle.

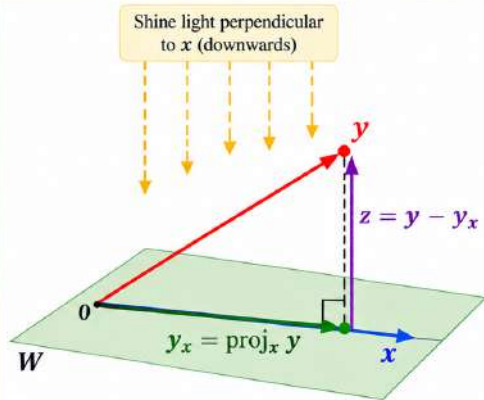
- If you **shine a light directly downwards** (perpendicular to $\mathbf{x}$), the shadow that $\mathbf{y}$ casts onto $\mathbf{x}$ is exactly the **orthogonal projection $\mathbf{y}_x$** .
- The projection lies perfectly flat along the line defined by $\mathbf{x}$.
- It is a scaled version of $\mathbf{x}$ (i.e., $\mathbf{y}_x = \alpha \mathbf{x}$).
- The "**gap**" between the tip of $\mathbf{y}$ and the tip of its shadow is the **orthogonal component $\mathbf{z}$** . This gap is perfectly perpendicular (90°) to $\mathbf{x}$.

2. Decomposition into Two Perpendicular Parts

The equation $\mathbf{y} = \mathbf{y}_x + \mathbf{z}$ splits any vector into two independent pieces:

- $\mathbf{y}_x$ (**The Projection**): The part of $\mathbf{y}$ that points in the same direction as $\mathbf{x}$.
- $\mathbf{z}$ (**The Orthogonal Component**): The part of $\mathbf{y}$ that points perpendicular to $\mathbf{x}$.

Together, these two vectors form a right-angled triangle, where $\mathbf{y}$ is the hypotenuse.



$$\mathbf{y} = \mathbf{y}_x + \mathbf{z}$$

$\mathbf{y}_x$ and $\mathbf{z}$ are perpendicular, forming a right-angled triangle with $\mathbf{y}$ as the hypotenuse.

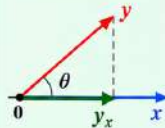
Geometric Interpretation of Orthogonal Projection

3. Positive vs. Negative Projection

The direction of the projection depends on the angle θ between the two vectors:

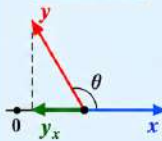
- If $\theta < 90^\circ$ (**Acute angle**): The projection points in the **same direction** as x . The scalar α is **positive**.
- If $\theta > 90^\circ$ (**Obtuse angle**): The projection points in the **opposite direction** to x . The scalar α is **negative** (the shadow falls behind x).
- If $\theta = 90^\circ$ (**Orthogonal**): The projection is the **zero vector**. The "shadow" is just a single point.

$\theta < 90^\circ$
(Acute angle)



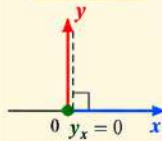
Projection in the same
direction as x
($\alpha > 0$)

$\theta > 90^\circ$
(Obtuse angle)



Projection in the opposite
direction to x
($\alpha < 0$)

$\theta = 90^\circ$
(Orthogonal)

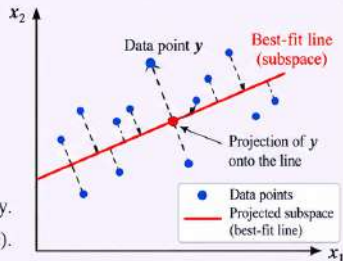


Projection is the zero vector
(the shadow is a single point)
($\alpha = 0$)

4. Applications in Machine Learning

Orthogonal projection answers: "How much of vector y goes in the direction of vector x ?"

- **Least Squares Regression:** Finding the best-fit line by projecting data points onto a subspace. This minimizes the sum of squared errors.
- **Principal Component Analysis (PCA):** Projecting high-dimensional data onto a lower-dimensional subspace that captures the maximum variance.
- **Gram-Schmidt Process:** Creating orthogonal bases for feature engineering and numerical stability.
- **Computer Graphics:** Calculating lighting and shading (how much light from a source hits a surface).



Orthogonal Projection

Given a nonzero vector $\mathbf{x}$, consider decomposing a vector $\mathbf{y}$ into the sum of two vectors: one a multiple of $\mathbf{x}$ and the other orthogonal to $\mathbf{x}$.

$$\mathbf{y} = \mathbf{y}_x + \mathbf{z} \quad (1)$$

where $\mathbf{y}_x = \alpha\mathbf{x}$ for some scalar α , and $\mathbf{z}$ is orthogonal to $\mathbf{x}$.

Derivation: Given any scalar α , let $\mathbf{z} = \mathbf{y} - \mathbf{y}_x$, so that (1) is satisfied. Then $\mathbf{y} - \mathbf{y}_x$ is orthogonal to $\mathbf{x}$ if and only if:

$$\begin{aligned} 0 &= (\mathbf{y} - \alpha\mathbf{x}) \cdot \mathbf{x} \\ &= \mathbf{y} \cdot \mathbf{x} - (\alpha\mathbf{x}) \cdot \mathbf{x} \\ &= \mathbf{y} \cdot \mathbf{x} - \alpha(\mathbf{x} \cdot \mathbf{x}) \end{aligned}$$

Orthogonal Projection

That is, (1) is satisfied with $\mathbf{z}$ orthogonal to $\mathbf{x}$ if and only if:

$$\alpha = \frac{\mathbf{y} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}}$$

Scalar Projection: The scalar projection (or component) of $\mathbf{y}$ onto $\mathbf{x}$ is the signed length of the projection:

$$\text{comp}_{\mathbf{x}}\mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{x}}{\|\mathbf{x}\|}$$

Vector Projection: The vector projection of $\mathbf{y}$ onto $\mathbf{x}$ is the orthogonal projection vector:

$$\text{proj}_{\mathbf{x}}\mathbf{y} = \mathbf{y}_{\mathbf{x}} = \frac{\mathbf{y} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}}\mathbf{x} = \left(\frac{\mathbf{y} \cdot \mathbf{x}}{\|\mathbf{x}\|} \right) \frac{\mathbf{x}}{\|\mathbf{x}\|} = \left(\frac{\mathbf{y} \cdot \mathbf{x}}{\|\mathbf{x}\|} \right) \hat{\mathbf{x}}$$

Orthogonal Projection

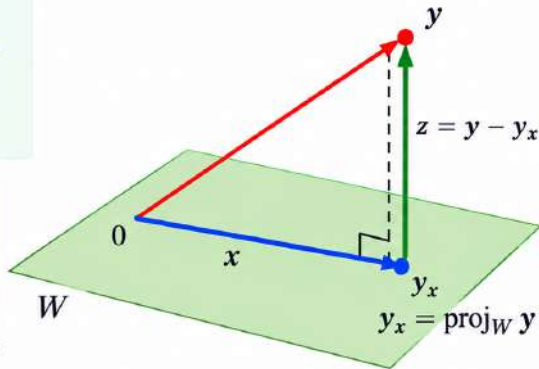
Consider two d -dimensional vectors $\mathbf{x} = [x_1, x_2, \dots, x_d]$ and $\mathbf{y} = [y_1, y_2, \dots, y_d]$:

Orthogonal Projection:

The vector $\mathbf{y}_x$ is called the **orthogonal projection** of $\mathbf{y}$ onto $\mathbf{x}$, and the vector $\mathbf{z}$ is called the **component** of $\mathbf{y}$ **orthogonal** to $\mathbf{x}$.

$$\mathbf{y}_x = \frac{\mathbf{y} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}} \mathbf{x} = \frac{\mathbf{y} \cdot \mathbf{x}}{\|\mathbf{x}\|^2} \hat{\mathbf{x}}$$

where $\mathbf{y} \cdot \mathbf{x}$ is the dot product and $\hat{\mathbf{x}}$ is the unit vector in the direction of $\mathbf{x}$.



Practice Exercises: Orthogonal Projection

Question 1:

Given vectors $\mathbf{a} = (3, 4)$ and $\mathbf{b} = (1, 2)$, find:

- (a) The scalar projection (component) of $\mathbf{b}$ onto $\mathbf{a}$.
- (b) The vector projection of $\mathbf{b}$ onto $\mathbf{a}$.

Question 2:

Let $\mathbf{p} = (2, -1, 4)$ and $\mathbf{q} = (-1, 3, 2)$. Calculate the vector projection of $\mathbf{q}$ onto $\mathbf{p}$.

Question 3:

After finding $\text{proj}_{\mathbf{a}}(\mathbf{b})$ for $\mathbf{a} = (2, 1)$ and $\mathbf{b} = (3, 4)$, verify that the vector $\mathbf{b} - \text{proj}_{\mathbf{a}}(\mathbf{b})$ is orthogonal to $\mathbf{a}$.

Equation of a Line in Two Variables

General (Linear) Form

$$ax + by + c = 0$$

where $a, b, c \in \mathbb{R}$ and $(a, b) \neq (0, 0)$.

Slope-Intercept Form

$$y = mx + b$$

- m is the slope
- b is the y -intercept (the point where the line crosses the y -axis)

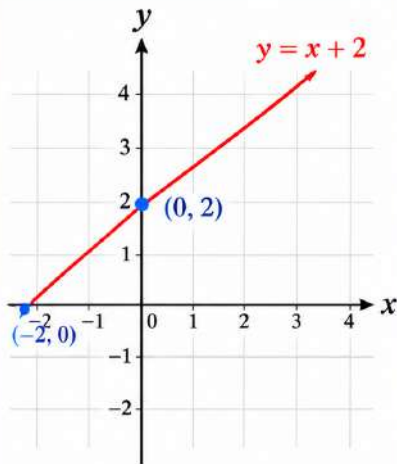
Example

The equation $2x - y + 4 = 0$ represents a line.

Solving for y , we get

$$y = 2x + 4$$

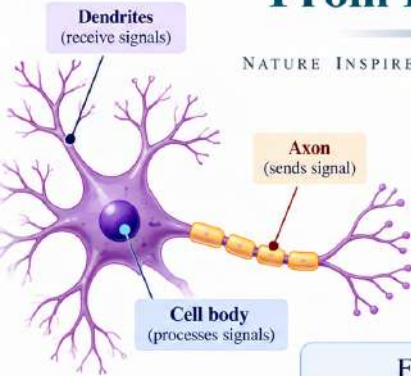
So the slope is $m = 2$ and the y -intercept is $b = 4$.



Unfolding the Neuron

From Biology to Mathematics

NATURE INSPIRES • MODELS CONNECT • LEARNING EMPOWERS



**Biological
Neuron**



**Artificial
Neuron
(Perceptron)**



**Learning
Algorithm**



From the brain's simple building block
to powerful machine learning models



THE HUMAN BRAIN - A MIRACLE

THE ULTIMATE SUPERCOMPUTER

Your brain is the most complex structure in the known universe.



~100 BILLION NEURONS

Each capable of thousands of connections.



~100 TRILLION CONNECTIONS

More than the number of stars in the Milky Way



< 20 WATTS OF POWER

Less power than a dim light bulb.

BEYOND STORAGE

Your brain can store more information than any library or device.



ESTIMATED STORAGE CAPACITY:

2.5 PETABYTES

(2.5 million gigabytes)



More than all the world's libraries combined.

THE HUMAN BRAIN - A MIRACLE

LIGHTNING FAST

Your brain processes information at incredible speed.



PROCESSES INFORMATION AT UP TO

**10^{17} OPERATIONS
PER SECOND**

Faster than the most powerful
supercomputers.

BUILDER OF YOU

It creates your memories, personality, thoughts, and consciousness.



Feels love and empathy



Solves problems and makes decisions



Imagines, dreams, and creates



Understands right from wrong

A Neuron: Building blocks of brain also referred to as brain cells.

Dendrites

(receive signals from other neurons)

Axon terminals

(release signals to other neurons)

Axon

(sends signal to other neurons)

Cell body

(processes information)

A neuron receives, processes, and transmits information through electrical and chemical signals.



“The human brain has **100 billion** neurons, each neuron is connected to **10 thousand** other neurons.”



~100 billion neurons



Each neuron connects to ~10 thousand others



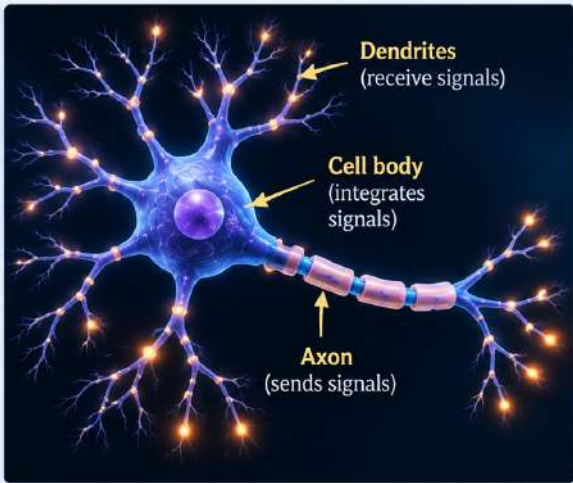
An intricate network for thinking, learning, and memory

Tiny cells. Massive possibilities.

REAL NEURON: HOW IT WORKS

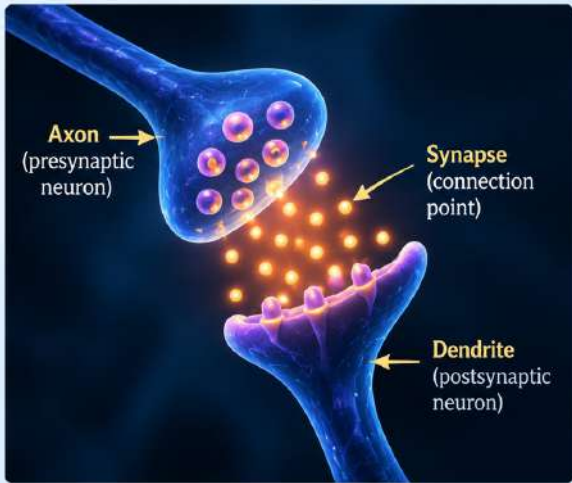
1

Neurons are connected to one another with the use of axons and dendrites.



2

The connecting regions between axons and dendrites are referred to as synapses.

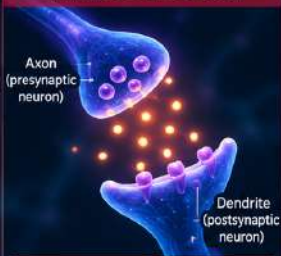


REAL NEURON: HOW IT WORKS

3

The strengths of synaptic connections often change in response to external stimuli.

WEAK CONNECTION



Fewer neurotransmitters released and fewer receptors activated.
Weaker signal.

STRONG CONNECTION

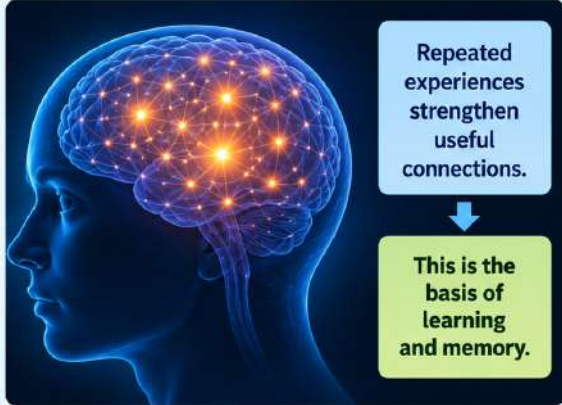


More neurotransmitters released and more receptors activated.
Stronger signal.

Synaptic strength can increase or decrease based on experience and activity.

4

This change is how learning takes place in living organisms.

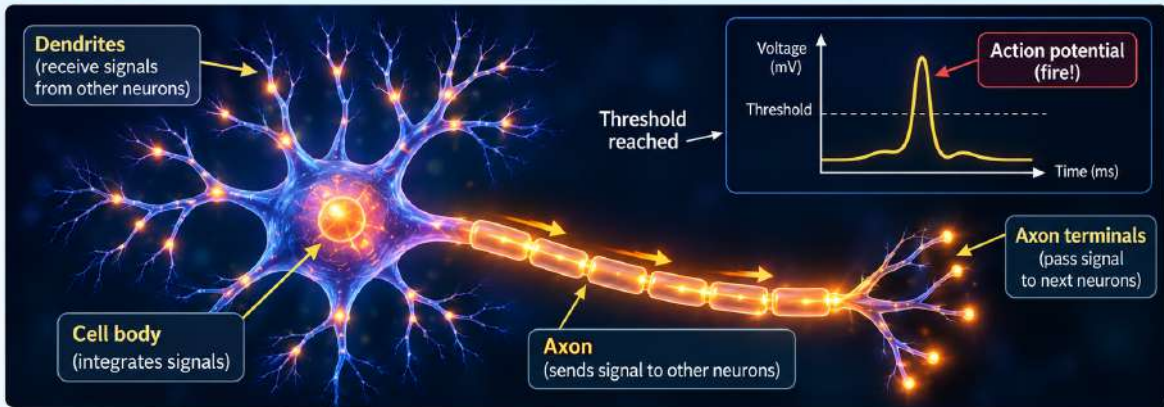


Our brains learn by changing the strength of connections between neurons.

REAL NEURON: HOW IT WORKS

6

If the resulting signal exceeds some threshold, then the neuron will “fire” a voltage impulse in response.



Neurons communicate. Synapses adapt. Signals integrate. When the threshold is crossed, the neuron fires. This is how the brain processes information, learns, and adapts.

ARTIFICIAL NEURON: THE THRESHOLD LOGIC UNIT

1

Inputs (signals)

Dendrites and axons are just communication channels.

2

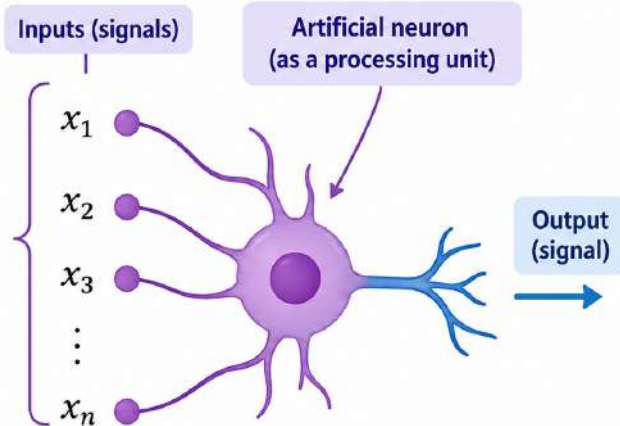
Mathematical representation

In math, we call these inputs—represented by the variables x .

3

Each input is a number

Each input is a number—like a pixel value, a word embedding, or a sensor reading.



The artificial neuron receives numerical inputs, processes them, and produces an output based on a threshold rule.

ARTIFICIAL NEURON: THE THRESHOLD LOGIC UNIT

2

Synaptic strengths (weights)

The synaptic strength is represented by a weight.

2.1

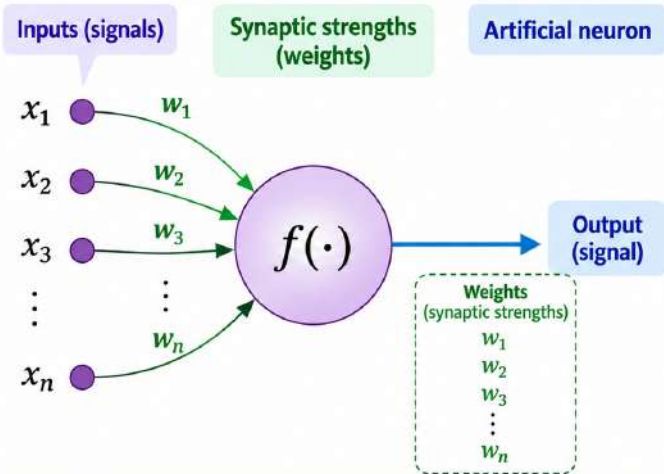
What is a weight?

A weight controls how strongly an input signal influences the neuron.

2.2

Positive or negative

A positive weight excites the neuron (increases its activation), while a negative weight inhibits it (decreases activation).

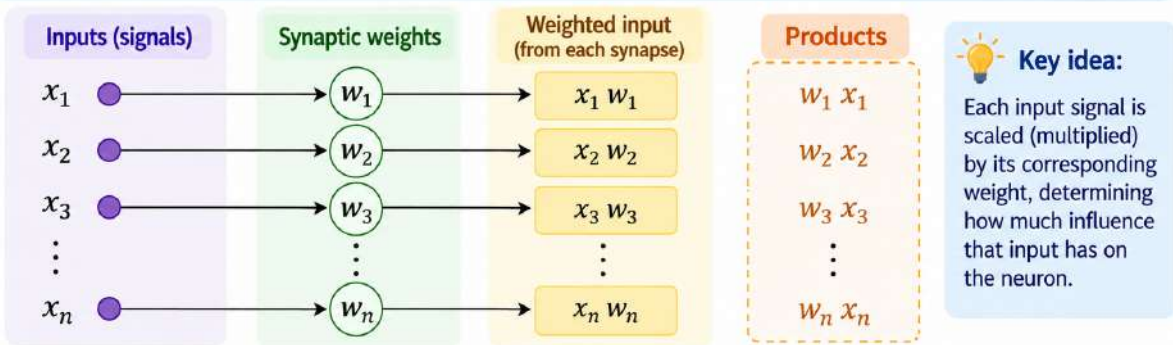


Each input x_i is **multiplied by a weight w_i** , which determines the **strength and direction** of its influence on the neuron.

ARTIFICIAL NEURON: THE THRESHOLD LOGIC UNIT

3 Effect of each synapse (multiplication)

The effect of each synapse is encapsulated by simply multiplying the incoming signal with a weight value w .



Each synapse multiplies the incoming signal by a weight value.
These weighted inputs are then combined in the next step.

ARTIFICIAL NEURON: **THE THRESHOLD LOGIC UNIT**

4 n weights and n products

- We therefore have n weights $w_1, w_2, \dots, w_n$
- and form the n products $w_1 x_1, w_2 x_2, \dots, w_n x_n$.
- Each product $w_i x_i$ represents the contribution of input x_i to the neuron's total input.

Weights
(synaptic strengths)

w_1

w_2

w_3

$\vdots$

w_n

=

=

=

=

=

Products
(weighted inputs)

$w_1 x_1$

$w_2 x_2$

$w_3 x_3$

$\vdots$

$w_n x_n$

Each input is multiplied by its corresponding weight to determine how much that input influences the neuron.

ARTIFICIAL NEURON: THE THRESHOLD LOGIC UNIT

5

Combine the products (summation)

All the weighted inputs are added together to produce the activation a .

5.1

Weighted sum

Each input x_i is multiplied by its corresponding weight w_i , and all products are added together.

5.2

Dot product form

The activation a is the dot product of the weight vector $\mathbf{w}$ and the input vector $\mathbf{x}$.

5.3

Vector notation

If $\mathbf{w} = [w_1, w_2, \dots, w_n]^T$ and $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$, then $a = \mathbf{w}^T \mathbf{x} = \sum_{i=1}^n w_i x_i$.

Input vector

$\mathbf{x}$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

Weight vector

$\mathbf{w}$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix}$$

$\cdot$

$=$

$$\begin{array}{c} \text{Element-wise products} \\ \hline w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n \\ \hline \text{Activation (a)} \\ \text{(sum of all products)} \end{array}$$

$$a = \mathbf{w}^T \mathbf{x} = \sum_{i=1}^n w_i x_i$$

The activation a is the dot product of the weight vector $\mathbf{w}$ and the input vector $\mathbf{x}$.

The weighted sum combines all inputs into a single value (the activation) that is then passed to the threshold (activation function).

ARTIFICIAL NEURON: THE THRESHOLD LOGIC UNIT

6 Activation (a)

All weighted inputs are added (dot product) to produce the net input or activation a .

$$a = w_1 x_1 + w_2 x_2 + w_3 x_3 + \cdots + w_n x_n \\ = \sum_{i=1}^n w_i x_i = \mathbf{w}^T \mathbf{x}$$

where

x_i = input signals

w_i = synaptic weights

a = activation (net input)

7 Apply activation function (threshold)

The activation is passed through a threshold function $f(a)$ to decide whether the neuron fires.

$$f(a) = \begin{cases} 1 & \text{if } a \geq \theta \\ 0 & \text{if } a < \theta \end{cases}$$

θ = threshold

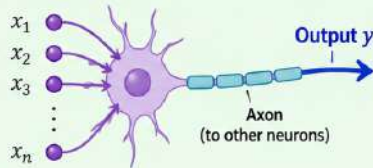
If the net input is greater than or equal to the threshold, the neuron is activated.

8 Output (action potential)

The **output** of the neuron is a binary signal (0 or 1) given by

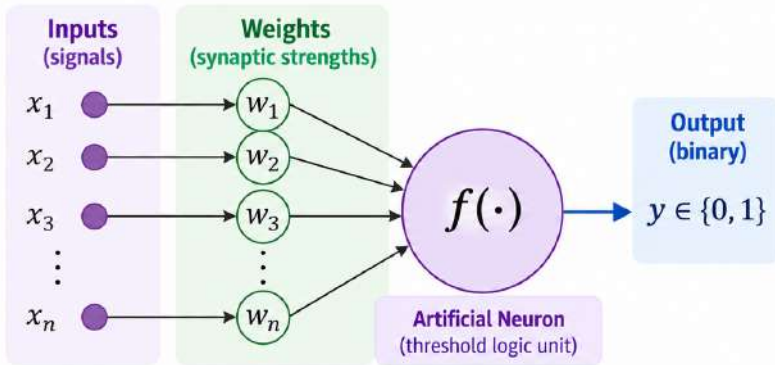
$$y = f(a) = \begin{cases} 1 & \text{(fire) if } a \geq \theta \\ 0 & \text{(no fire) if } a < \theta \end{cases}$$

This output (action potential) is sent through the axon to other neurons.



The neuron computes a weighted sum of its inputs and applies a threshold function to produce a binary output (0 or 1).

Summary: Artificial Neuron (McCulloch & Pitts, 1943)



1 Weighted sum (net input)

$$a = \sum_{i=1}^n w_i x_i = \mathbf{w}^T \mathbf{x}$$

2 Activation function (threshold)

$$f(a) = \begin{cases} 1 & \text{if } a \geq \theta \\ 0 & \text{if } a < \theta \end{cases}$$

$\theta = \text{threshold}$

3 Output (action potential)

$$y = f(a) \in \{0, 1\}$$

This output is sent through the axon to other neurons.

Key points:

- Each input x_i is multiplied by a weight w_i .
- The weighted inputs are summed to give the net input a .
- A threshold function $f(a)$ decides whether the neuron fires.
- The output is a binary signal (0 or 1).

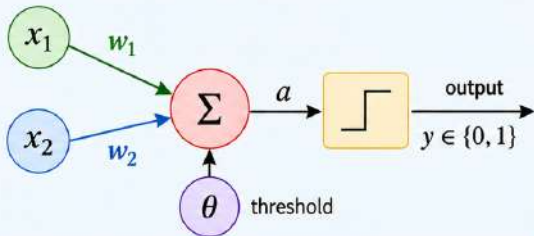
Historical significance:

- Introduced by McCulloch & Pitts (1943).
- A simple mathematical model of a neuron.
- Foundation for modern artificial neural networks.

THE LINEAR SEPARATION OF CLASSES

1 Perceptron with Two Inputs

Consider a perceptron with two inputs x_1 and x_2 , weights w_1 and w_2 , activation a and threshold θ .



Activation (net input):

$$a = w_1 x_1 + w_2 x_2$$

Output (threshold function):

$$y = f(a) = \begin{cases} 1 & \text{if } a \geq \theta \\ 0 & \text{if } a < \theta \end{cases}$$

2 Critical Condition (Decision Boundary)

The critical condition for classification occurs when the activation equals the threshold.

$$a = \theta$$

So,

$$w_1 x_1 + w_2 x_2 = \theta$$

This is the equation of the decision boundary that separates the two classes.

THE LINEAR SEPARATION OF CLASSES

3 Manipulate Mathematically

Rearrange the equation:

$$w_1 x_1 + w_2 x_2 = \theta$$

Solve for x_2 , assuming $w_2 \neq 0$:

$$w_2 x_2 = \theta - w_1 x_1$$

$$x_2 = -\frac{w_1}{w_2} x_1 + \frac{\theta}{w_2}$$

4 General Form of a Line

This is of the general form:

$$x_2 = a x_1 + b$$

where

$$a = -\frac{w_1}{w_2} \quad (\text{slope})$$

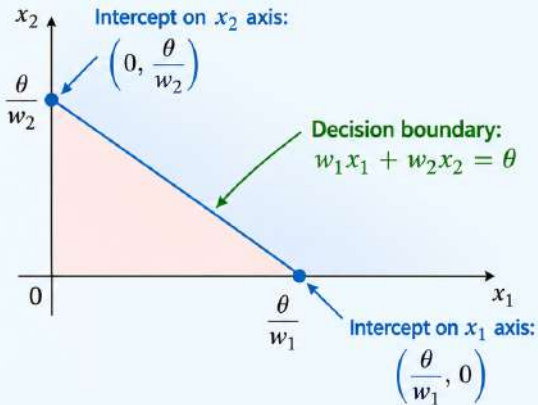
$$b = \frac{\theta}{w_2} \quad (\text{intercept on } x_2 \text{ axis})$$

This equation describes a straight line with slope a and intercept b on the x_2 axis.

THE LINEAR SEPARATION OF CLASSES

5 Geometric Interpretation

The line $w_1x_1 + w_2x_2 = \theta$ divides the x_1, x_2 plane into two half-planes.



6 Classification Regions

Which side is class 1 or class 0 depends on the activation and threshold.

If

$$a = w_1x_1 + w_2x_2 \geq \theta$$

then

$$y = 1$$

If

$$a = w_1x_1 + w_2x_2 < \theta$$

then

$$y = 0$$

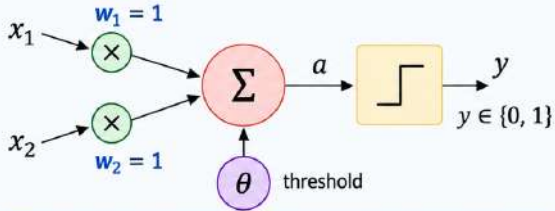
The line itself ($a = \theta$) is the boundary.

LINEAR SEPARATION OF CLASSES (EXAMPLE)

1 Step 1: The Two-Input TLU

Consider a two-input TLU with:

$$w_1 = 1, \quad w_2 = 1, \quad \theta = 1.5$$



Activation (net input): $a = w_1x_1 + w_2x_2$
(weighted sum)

Output (threshold function):

$$y = \begin{cases} 1 & \text{if } a \geq \theta \\ 0 & \text{if } a < \theta \end{cases}$$

2 Step 2: Compute Activation for All Input Patterns

x_1	x_2	Activation $a = x_1 + x_2$	Threshold $\theta = 1.5$	Output y
0	0	0	1.5	0
0	1	1	1.5	0
1	0	1	1.5	0
1	1	2	1.5	1

Conclusion:

The TLU gives output 1 only when the activation is greater than or equal to 1.5.

LINEAR SEPARATION OF CLASSES (EXAMPLE)

3 Step 3: Decision Boundary Equation

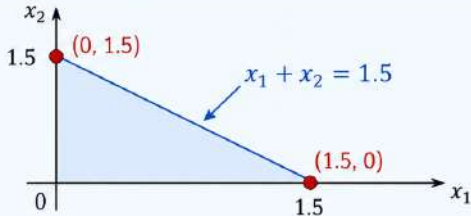
The decision boundary occurs when $a = \theta$.

$$\begin{aligned}w_1 x_1 + w_2 x_2 &= \theta \\1 \cdot x_1 + 1 \cdot x_2 &= 1.5 \\x_1 + x_2 &= 1.5\end{aligned}$$

This is the equation of a straight line that separates the input space.

Solve for x_2 :

$$\begin{aligned}x_2 &= -x_1 + 1.5 \\(\text{slope} &= -1, \text{intercept on } x_2 \text{ axis} = 1.5)\end{aligned}$$



2 Step 2: Compute Activation for All Input Patterns

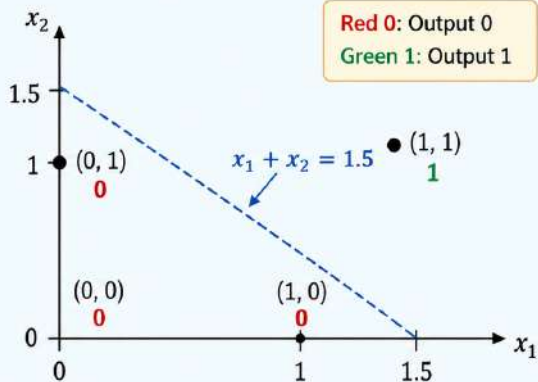
x_1	x_2	Activation $a = x_1 + x_2$	Threshold $\theta = 1.5$	Output y
0	0	0	1.5	0
0	1	1	1.5	0
1	0	1	1.5	0
1	1	2	1.5	1

The TLU gives output 1 only when the activation is greater than or equal to 1.5.

LINEAR SEPARATION OF CLASSES (EXAMPLE)

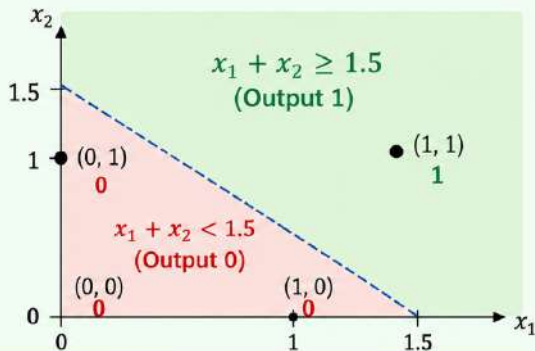
4 Step 4: Plot the Input Patterns

Plot the four input patterns and the decision boundary given by $x_1 + x_2 = 1.5$.



5 Step 5: Classification Regions

The decision boundary divides the input space into two regions.



Points with $x_1 + x_2 \geq 1.5$ are classified as 1, and points with $x_1 + x_2 < 1.5$ are classified as 0.

Training TLU: The Perceptron Rule

$$\mathbf{w} \cdot \mathbf{x} \geq \theta$$

$$\mathbf{w} \cdot \mathbf{x} - \theta \geq 0$$

$$[w_1, w_2, \dots, w_n, \theta] \cdot [x_1, x_2, \dots, -1] \geq 0$$

TLU may be expressed as the output function as follows:

$$y = \begin{cases} 1, & \text{if } \mathbf{w} \cdot \mathbf{x} \geq 0 \\ 0, & \text{if } \mathbf{w} \cdot \mathbf{x} < 0 \end{cases}$$

$\mathbf{w} \cdot \mathbf{x} = 0$ now defines the decision hyper-plane.

Training TLU: The Perceptron Rule

Example: Consider two input TLU with weight vector $\mathbf{w} = [\frac{1}{2}, \frac{1}{2}]$ and $\theta = \frac{3}{4} = 0.75$.

x_1	x_2	$a = w_1x_1 + w_2x_2$	y
0	0		
0	1		
1	0		
1	1		

Exercise: Explain the decision line in 2D?

Training TLU: The Perceptron Rule

Example: Consider two input TLU with weight vector $\mathbf{w} = [\frac{1}{2}, \frac{1}{2}]$ and $\theta = \frac{3}{4} = 0.75$.

x_1	x_2	$a = w_1x_1 + w_2x_2$	y
0	0	0	0
0	1	0.5	0
1	0	0.5	0
1	1	1	1

Exercise: Explain the decision line in 2D?

$$\mathbf{w} \cdot \mathbf{x} = \theta$$

$$\frac{1}{2}x_1 + \frac{1}{2}x_2 = \frac{3}{4} \implies x_1 + x_2 = \frac{3}{2} = 1.5$$

Training TLU: The Perceptron Rule

- ▶ For augmented pattern space, we have to consider 3-Dimension.
- ▶ The two components x_1, x_2 are now drawn in the horizontal plane while the third component x_3 has been introduced as the third axis, i.e., $(x_1, x_2, -1)$.
- ▶ The augmented weight vector $\mathbf{w} = [\frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$.
- ▶ The old decision line in 2D is formed by the intersection of the decision plane and the plane containing the patterns.

Exercise:

Q:1 Find the equation of the plane containing the patterns.

Q:2 Find the equation of the decision plane.

Q:3 Graph these two planes in 3D and observe the intersection as the decision line.

▶ [Click here to open the 3D GeoGebra Visualization](#)